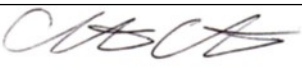


Guideline Title: Cricothyrotomy - Surgical

Guideline Number: GUID-3203-PR007

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

I. **PURPOSE:** To allow rapid entrance into the airway for ventilation and oxygenation when other means of airway control (Bag Valve Mask, intubation, etc.) have proven unsuccessful.

II. **DEFINITIONS:**

a. ETCO₂ – End tidal carbon dioxide

III. **DEPARTMENT GUIDELINE:**

a. **EQUIPMENT:**

- i. Oxygen
- ii. Suction
- iii. Bag Valve mask
- iv. Using the H&H Emergency Cricothyrotomy Kit
 1. Contents of kit includes: #10 scalpel, 10 ml syringe, 35 F Bougie, Tracheal hook, Kelly forceps/clamp, Short 6.0 Airway catheter (trach) with flange, Tracheal ties
 2. Additional supplies needed: 4 x 4's, Chlorohexidine or Betadine cleaning solution
- v. For standard surgical technique
 1. #10 Scalpel
 2. 6.0 – 7.0 ETT
 3. Chlorohexidine or Betadine cleaning solution
 4. Gauze
 5. Kelly forceps/clamp

IV. **INDICATIONS:**

- a. Inability to oxygenate or ventilate the patient by any other means including subglottic airway
- b. Total airway obstruction
- c. Severe trauma compromising airway

V. **CONTRAINDICATIONS:**

- a. The ability to obtain airway control and effective ventilation by less invasive means.
- b. Pediatric patients (< 8 years old)
- c. Inability to identify proper landmarks.

Guideline Title: Cricothyrotomy - Surgical

Guideline Number: GUID-3203-PR007

VI. PROCEDURES:

a. Using the H&H Emergency Cricothyroid Kit

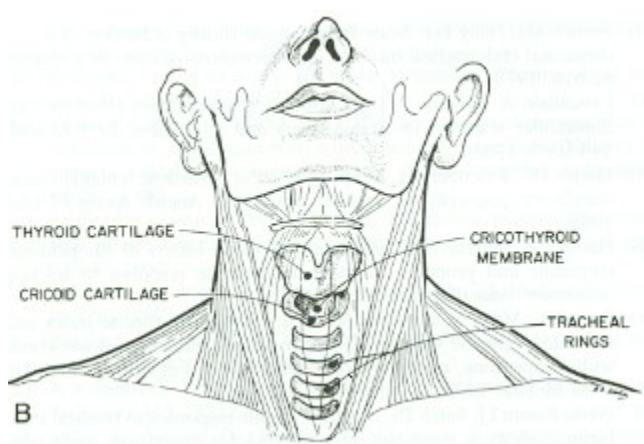
- i. Open the airway and position the head so the neck is clearly visible. If the patient has sustained any type of spinal trauma, always maintain cervical spine precautions.
- ii. After locating and palpating the cricothyroid membrane, consider marking with a pen.
- iii. Clean the area with Betadine or chlorohexidine.
- iv. Attach the 10 ml syringe to the balloon port. Ensure bougie is through the tracheal tube.
- v. Stabilize the cricothyroid membrane and make a 2-3 cm vertical incision in the midline using the #10 scalpel blade. The incision should be long enough to accommodate the dilator and trach. Dissect bluntly through the subcutaneous tissues until the membrane is visible.
- vi. Make a second 1.5-2 cm horizontal incision through the cricoid membrane using the scalpel blade.
- vii. If bleeding occurs, control with forceps and gauze. Suction area as needed.
- viii. Place the hook into the hole and apply mild traction towards the head of the patient.
- ix. Introduce the bougie until you feel the resistance of the carina.
- x. Remove the hook and introduce the tracheal tube.
- xi. Remove the bougie.
- xii. Inflate the cuff.
- xiii. Manually secure the tracheostomy tube while beginning to ventilate the patient using a Bag Valve device with ETCO2 waveform capnography attached.
- xiv. Confirm placement by auscultating for equal, bilateral breath sounds and observing for equal, bilateral chest expansion.
- xv. Secure the tracheostomy tube in place with tracheostomy ties.

b. Using Standard Surgical Technique

- i. Open the airway and position the head so the neck is clearly visible. If the patient has sustained any type of spinal trauma, always maintain cervical spine precautions
- ii. After locating and palpating the cricothyroid membrane, consider marking with a pen.
- iii. Clean the area thoroughly with chlorohexidine or Betadine.
- iv. Stabilize the cricothyroid membrane and make a 2cm vertical incision in the midline using the #10 scalpel blade. Dissect bluntly through the subcutaneous tissues until the membrane is visible.
- v. Make a second 1.5-2 cm horizontal incision through the cricoid membrane using the scalpel blade.
- vi. Use a curved hemostat to apply traction to the cricothyroid membrane allowing for tube insertion.
- vii. Introduce a bougie until you feel the resistance of the carina.
- viii. Insert an appropriately sized, cuffed endotracheal tube into the cricothyroid membrane incision, directing the tube distally into the trachea until the cuff is securely in the trachea
- ix. Inflate the cuff.

Guideline Title: Cricothyrotomy - Surgical**Guideline Number: GUID-3203-PR007**

- x. Manually secure the tube while beginning to ventilate the patient using a bag valve device with ETCO2 waveform capnography attached.
 - xi. Confirm placement by auscultating for equal, bilateral breath sounds and observing for equal, bilateral chest expansion.
 - xii. Secure tube with cloth tape if available
- c. Illustration of Tracheal Anatomy



VII. **DOCUMENTATION:** EMS Charts

VIII. **REFERENCES:**